

**Sequential Transfer Learning for Medicinal Plant Identification: Endangered
Species Protection and Sustainable Harvesting**

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APPENDIX A: USER GUIDE

Installation

Prerequisites:

- Python 3.9 or later
- pip package manager

Setup:

1. Extract the project ZIP archive.
2. Navigate to the project directory:
`cd medicinal-plant-identification`
3. Create a virtual environment:
`python -m venv .venv`
4. Activate the virtual environment:
`.venv\Scripts\Activate.ps1` (Windows PowerShell)
`source .venv/bin/activate` (Linux/macOS)
5. Install dependencies:
`pip install -r requirements.txt`
6. Ensure the trained model file is placed in `webapp/models/`

Running the Web Application:

```
-----  
cd webapp  
python app.py  
-----
```

The application will start on `http://localhost:5000`.

Using the Application:

1. Open a web browser and navigate to `http://localhost:5000`.
2. Either capture an image using the camera button or upload a leaf image.
3. The system will display the top-5 predicted species with confidence scores, enhanced by test-time augmentation.
4. Out-of-distribution detection will flag non-plant or ambiguous images.
5. Click on a species name to view detailed information including conservation status, harvesting legality, and medicinal uses.
6. View prediction history at `/history`.